US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12389667
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Tsai; Meng Jhe et al.

Fin field-effect transistor device and method

Abstract

A method of forming a semiconductor device includes: forming a metal gate structure over a fin that protrudes above a substrate, the metal gate structure being surrounded by an interlayer dielectric (ILD) layer; recessing the metal gate structure below an upper surface of the ILD layer distal from the substrate; after the recessing, forming a first dielectric layer over the recessed metal gate structure; forming an etch stop layer (ESL) over the first dielectric layer and the ILD layer; forming a second dielectric layer over the ESL; performing a first dry etch process to form an opening that extends through the second dielectric layer, through the ESL, and into the first dielectric layer; after the first dry etch process, performing a wet etch process to clean the opening; and after the wet etch process, performing a second dry etch process to extend the opening through the first dielectric layer.

Inventors: **Tsai; Meng Jhe (Hsinchu, TW), Yang; Hong-Jie (Hsinchu, TW), Chang; Meng-Chun (Taipei, TW), Chiang; Hao (Hsinchu, TW), Lee; Chia-Ying (New Taipei, TW), Lin; Huan-Just (Hsinchu, TW), Chang; Chuan (Taipei, TW)**

Applicant: **Taiwan Semiconductor Manufacturing Co., Ltd.** (Hsinchu, TW)

Family ID: **82235309**

Assignee: **Taiwan Semiconductor Manufacturing Co., Ltd. (Hsinchu, TW)**

Appl. No.: **18/741063**

Filed: **June 12, 2024**

Prior Publication Data

Document Identifier	Publication Date
US 20240332085 A1	Oct. 03, 2024

Related U.S. Application Data

continuation parent-doc US 17329998 20210525 US 12040233 child-doc US 18741063
us-provisional-application US 63159008 20210310

Publication Classification

Int. Cl.: **H10D30/62** (20250101); **H10D30/01** (20250101); **H10D84/01** (20250101); **H10D84/03** (20250101); **H10D84/85** (20250101)

U.S. Cl.:

CPC **H10D84/038** (20250101); **H10D30/024** (20250101); **H10D30/6211** (20250101); **H10D84/0158** (20250101); **H10D84/853** (20250101);

Field of Classification Search

CPC: H01L (21/823431); H01L (27/0924); H01L (29/66795); H01L (29/7851); H10D (30/024); H10D (30/6211); H10D (30/025); H10D (30/027); H10D (30/031); H10D (30/0312); H10D (30/0314); H10D (30/0316); H10D (30/6212); H10D (30/67); H10D (30/6728)

USPC: 438/478

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8207060	12/2011	Yang	N/A	N/A
8828881	12/2013	Meng et al.	N/A	N/A
9105490	12/2014	Wang et al.	N/A	N/A
9236267	12/2015	De et al.	N/A	N/A
9236300	12/2015	Liaw	N/A	N/A
9406804	12/2015	Huang et al.	N/A	N/A
9443769	12/2015	Wang et al.	N/A	N/A
9520482	12/2015	Chang et al.	N/A	N/A
9548366	12/2016	Ho et al.	N/A	N/A
9576814	12/2016	Wu et al.	N/A	N/A
9831183	12/2016	Lin et al.	N/A	N/A
9859386	12/2017	Ho et al.	N/A	N/A
10163704	12/2017	Chen et al.	N/A	N/A
10249530	12/2018	Wang et al.	N/A	N/A
10522640	12/2018	Jangjian et al.	N/A	N/A
10714329	12/2019	Lin et al.	N/A	N/A
12040233	12/2023	Tsai	N/A	H01L 29/66795
2015/0228793	12/2014	Chen et al.	N/A	N/A
2016/0181163	12/2015	Huang et al.	N/A	N/A
2018/0174904	12/2017	Hsieh et al.	N/A	N/A
2020/0006228	12/2019	Yang et al.	N/A	N/A
2020/0083379	12/2019	Chang et al.	N/A	N/A
2020/0105931	12/2019	Wu et al.	N/A	N/A
2020/0365698	12/2019	Tsai et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
101903990	12/2009	CN	N/A
20160129664	12/2015	KR	N/A
20170078514	12/2016	KR	N/A
20200037071	12/2019	KR	N/A
201714217	12/2016	TW	N/A
2012092695	12/2011	WO	N/A

Primary Examiner: Harrison; Monica D

Attorney, Agent or Firm: Slater Matsil, LLP

Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE (1) This application is a continuation of U.S. patent application Ser. No. 17/329,998, filed May 25, 2021, entitled “Fin Field-Effect Transistor Device and Method,” which claims priority to U.S. Provisional Patent Application No. 63/159,008, filed Mar. 10, 2021, entitled “Optimized Etch Sequence for VG Etch Beyond 3 nm Node,” which applications are hereby incorporated by reference in their entireties.

BACKGROUND
(1) The semiconductor industry has experienced rapid growth due to continuous improvements in the integration density of a variety of electronic components (e.g., transistors, diodes, resistors, capacitors, etc.). For the most part, this improvement in integration density has come from repeated reductions in minimum feature size, which allows more components to be integrated into a given area.
(2) Fin Field-Effect Transistor (FinFET) devices are becoming commonly used in integrated circuits. FinFET devices have a three-dimensional structure that comprises a semiconductor fin protruding from a substrate. A gate structure, configured to control the flow of charge carriers within a conductive channel of the FinFET device, wraps around the semiconductor fin. For example, in a tri-gate FinFET device, the gate structure wraps around three sides of the semiconductor fin, thereby forming conductive channels on three sides of the semiconductor fin.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
(2) FIG. **1** is a perspective view of a Fin Field-Effect Transistor (FinFET), in accordance with some embodiments.
(3) FIGS. **2-24** illustrate cross-sectional views of a FinFET device at various stages of fabrication, in accordance with an embodiment.
(4) FIG. **25** illustrates a flow chart of a method of forming a semiconductor device, in accordance with some embodiments.
DETAILED DESCRIPTION

(5) The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact.
(6) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
(7) Embodiments of the present disclosure are discussed in the context of forming a semiconductor device, and in particular, in the context of forming vias for a Fin Field-Effect Transistor (FinFET) device. The principle of the disclosed embodiments may also be applied to other types of devices, such as planar devices.
(8) In accordance with an embodiment of the present disclosure, a multi-step etching process, which includes a first dry etch process, a wet etch process, and a second dry etch process, is performed to form via holes that extend through multiple dielectric layers (e.g., an oxide layer over a nitride layer) to expose the underlying conductive features. The multi-step etching process is advantageous for scenarios where residue metal regions are formed between the multiple dielectric layers, which residue metal regions are formed by insufficient removal of fill metals by a planarization process (e.g., CMP). Since the residue metal regions may block the via hole etching process, the disclosed multi-step etching process ensures that the via holes are formed properly, regardless of whether the residue metal regions exist.
(9) FIG. **1** illustrates an example of a FinFET **30** in a perspective view. The FinFET **30** includes a substrate **50** and a fin **64** protruding above the substrate **50**. Isolation regions **62** are formed on opposing sides of the fin **64**, with the fin **64** protruding above the isolation regions **62**. A gate dielectric **66** is along sidewalls and over a top surface of the fin **64**, and a gate electrode **68** is over the gate dielectric **66**. Source/drain regions **80** are in the fin **64** and on opposing sides of the gate dielectric **66** and the gate electrode **68**. FIG. **1** further illustrates reference cross-sections that are used in later figures. Cross-section B-B extends along a longitudinal axis of the gate electrode **68** of the FinFET **30**. Cross-section A-A is perpendicular to cross-section B-B and is along a longitudinal axis of the fin **64** and in a direction of, for example, a current flow between the source/drain regions **80**. Subsequent figures refer to these reference cross-sections for clarity.
(10) FIGS. **2-24** illustrate cross-sectional views of a FinFET device **100** at various stages of fabrication, in accordance with an embodiment. The FinFET device **100** is similar to the FinFET **30** in FIG. **1**, except for multiple fins and multiple gate structures. FIGS. **2-5** illustrate cross-sectional views of the FinFET device **100** along cross-section B-B, and FIGS. **6-24** illustrate cross-sectional views of the FinFET device **100** along cross-section A-A.
(11) FIG. **2** illustrates a cross-sectional view of a substrate **50**. The substrate **50** may be a semiconductor substrate, such as a bulk semiconductor, a semiconductor-on-insulator (SOI) substrate, or the like, which may be doped (e.g., with a p-type or an n-type dopant) or undoped. The substrate **50** may be a wafer, such as a silicon wafer. Generally, an SOI substrate includes a layer of a semiconductor material formed on an insulator layer. The insulator layer may be, for example, a buried oxide (BOX) layer, a silicon oxide layer, or the like. The insulator layer is provided on a substrate, typically a silicon substrate or a glass substrate. Other substrates, such as a multi-layered

or gradient substrate may also be used. In some embodiments, the semiconductor material of the substrate **50** may include silicon; germanium; a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof.

(12) Referring to FIG. **3**, the substrate **50** shown in FIG. **2** is patterned using, for example, photolithography and etching techniques. For example, a mask layer, such as a pad oxide layer **52** and an overlying pad nitride layer **56**, is formed over the substrate **50**. The pad oxide layer **52** may be a thin film comprising silicon oxide formed, for example, using a thermal oxidation process. The pad oxide layer **52** may act as an adhesion layer between the substrate **50** and the overlying pad nitride layer **56** and may act as an etch stop layer for etching the pad nitride layer **56**. In some embodiments, the pad nitride layer **56** is formed of silicon nitride, silicon oxynitride, silicon carbonitride, the like, or a combination thereof, and may be formed using low-pressure chemical vapor deposition (LPCVD) or plasma enhanced chemical vapor deposition (PECVD), as examples.

(13) The mask layer may be patterned using photolithography techniques. Generally, photolithography techniques utilize a photoresist material (not shown) that is deposited, irradiated (exposed), and developed to remove a portion of the photoresist material. The remaining photoresist material protects the underlying material, such as the mask layer in this example, from subsequent processing steps, such as etching. In this example, the photoresist material is used to pattern the pad oxide layer **52** and pad nitride layer **56** to form a patterned mask **58**, as illustrated in FIG. **3**.

(14) The patterned mask **58** is subsequently used to pattern exposed portions of the substrate **50** to form trenches **61**, thereby defining semiconductor fins **64** between adjacent trenches **61** as illustrated in FIG. **3**. In some embodiments, the semiconductor fins **64** are formed by etching trenches in the substrate **50** using, for example, reactive ion etch (RIE), neutral beam etch (NBE), the like, or a combination thereof. The etch may be anisotropic. In some embodiments, the trenches **61** may be strips (viewed from the top) parallel to each other, and closely spaced with respect to each other. In some embodiments, the trenches **61** may be continuous and surround the semiconductor fins **64**. The semiconductor fins **64** may also be referred to as fins **64** hereinafter.

(15) The fins **64** may be patterned by any suitable method. For example, the fins **64** may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers, or mandrels, may then be used to pattern the fins.

(16) FIG. **4** illustrates the formation of an insulation material between neighboring semiconductor fins **64** to form isolation regions **62**. The insulation material may be an oxide, such as silicon oxide, a nitride, the like, or a combination thereof, and may be formed by a high density plasma chemical vapor deposition (HDP-CVD), a flowable CVD (FCVD) (e.g., a CVD-based material deposition in a remote plasma system and post curing to make it convert to another material, such as an oxide), the like, or a combination thereof. Other insulation materials and/or other formation processes may be used. In the illustrated embodiment, the insulation material is silicon oxide formed by a FCVD process. An anneal process may be performed once the insulation material is formed. A planarization process, such as a chemical mechanical polish (CMP), may remove any excess insulation material and form top surfaces of the isolation regions **62** and top surfaces of the semiconductor fins **64** that are coplanar (not shown). The patterned mask **58** (see FIG. **3**) may also be removed by the planarization process.

(17) In some embodiments, the isolation regions **62** include a liner, e.g., a liner oxide (not shown), at the interface between the isolation region **62** and the substrate **50**/semiconductor fins **64**. In some embodiments, the liner oxide is formed to reduce crystalline defects at the interface between the substrate **50** and the isolation region **62**. Similarly, the liner oxide may also be used to reduce crystalline defects at the interface between the semiconductor fins **64** and the isolation region **62**. The liner oxide (e.g., silicon oxide) may be a thermal oxide formed through a thermal oxidation of a surface layer of substrate **50**, although other suitable method may also be used to form the liner oxide.
(18) Next, the isolation regions **62** are recessed to form shallow trench isolation (STI) regions **62**. The isolation regions **62** are recessed such that the upper portions of the semiconductor fins **64** protrude from between neighboring STI regions **62**. The top surfaces of the STI regions **62** may have a flat surface (as illustrated), a convex surface, a concave surface (such as dishing), or a combination thereof. The top surfaces of the STI regions **62** may be formed flat, convex, and/or concave by an appropriate etch. The isolation regions **62** may be recessed using an acceptable etching process, such as one that is selective to the material of the isolation regions **62**. For example, a dry etch, or a wet etch using dilute hydrofluoric (dHF) acid, may be performed to recess the isolation regions **62**.
(19) FIGS. **2** through **4** illustrate an embodiment of forming fins **64**, but fins may be formed in various different processes. For example, a top portion of the substrate **50** may be replaced by a suitable material, such as an epitaxial material suitable for an intended type (e.g., n-type or p-type) of semiconductor devices to be formed. Thereafter, the substrate **50**, with epitaxial material on top, is patterned to form semiconductor fins **64** that comprise the epitaxial material.
(20) As another example, a dielectric layer can be formed over a top surface of a substrate; trenches can be etched through the dielectric layer; homoepitaxial structures can be epitaxially grown in the trenches; and the dielectric layer can be recessed such that the homoepitaxial structures protrude from the dielectric layer to form fins.
(21) In yet another example, a dielectric layer can be formed over a top surface of a substrate; trenches can be etched through the dielectric layer; heteroepitaxial structures can be epitaxially grown in the trenches using a material different from the substrate; and the dielectric layer can be recessed such that the heteroepitaxial structures protrude from the dielectric layer to form fins.
(22) In embodiments where epitaxial material(s) or epitaxial structures (e.g., the heteroepitaxial structures or the homoepitaxial structures) are grown, the grown material(s) or structures may be in situ doped during growth, which may obviate prior and subsequent implantations although in situ and implantation doping may be used together. Still further, it may be advantageous to epitaxially grow a material in an NMOS region different from the material in a PMOS region. In various embodiments, the fins **64** may comprise silicon germanium (Si.sub.xGe.sub.1-x, where x can be between 0 and 1), silicon carbide, pure or substantially pure germanium, a III-V compound semiconductor, a II-VI compound semiconductor, or the like. For example, the available materials for forming III-V compound semiconductor include, but are not limited to, InAs, AlAs, GaAs, InP, GaN, InGaAs, InAlAs, GaSb, AlSb, AIP, GaP, and the like.
(23) FIG. **5** illustrates the formation of dummy gate structure **75** over the semiconductor fins **64**. Dummy gate structure **75** includes gate dielectric **66** and gate electrode **68**, in some embodiments. A mask **70** may be formed over the dummy gate structure **75**. To form the dummy gate structure **75**, a dielectric layer is formed on the semiconductor fins **64**. The dielectric layer may be, for example, silicon oxide, silicon nitride, multilayers thereof, or the like, and may be deposited or thermally grown.
(24) A gate layer is formed over the dielectric layer, and a mask layer is formed over the gate layer. The gate layer may be deposited over the dielectric layer and then planarized, such as by a CMP. The mask layer may be deposited over the gate layer. The gate layer may be formed of, for example, polysilicon, although other materials may also be used. The mask layer may be formed

of, for example, silicon nitride or the like.

(25) After the layers (e.g., the dielectric layer, the gate layer, and the mask layer) are formed, the mask layer may be patterned using acceptable photolithography and etching techniques to form mask **70**. The pattern of the mask **70** then may be transferred to the gate layer and the dielectric layer by an acceptable etching technique to form gate electrode **68** and gate dielectric **66**, respectively. The gate electrode **68** and the gate dielectric **66** cover respective channel regions of the semiconductor fins **64**. The gate electrode **68** may also have a lengthwise direction substantially perpendicular to the lengthwise direction of respective semiconductor fins **64**.

(26) The gate dielectric **66** is shown to be formed over the fins **64** (e.g., over top surfaces and sidewalls of the fins **64**) and over the STI regions **62** in the example of FIG. **5**. In other embodiments, the gate dielectric **66** may be formed by, e.g., thermal oxidization of a material of the fins **64**, and therefore, may be formed over the fins **64** but not over the STI regions **62**. These and other variations are fully intended to be included within the scope of the present disclosure.

(27) Next, as illustrated in FIG. **6**, lightly doped drain (LDD) regions **65** are formed in the fins **64**. The LDD regions **65** may be formed by an implantation process. The implantation process may implant n-type or p-type impurities in the fins **64** to form the LDD regions **65**. In some embodiments, the LDD regions **65** abut the channel region of the FinFET device **100**. Portions of the LDD regions **65** may extend under gate electrode **68** and into the channel region of the FinFET device **100**. FIG. **6** illustrates a non-limiting example of the LDD regions **65**. Other configurations, shapes, and formation methods of the LDD regions **65** are also possible and are fully intended to be included within the scope of the present disclosure. For example, LDD regions **65** may be formed after gate spacers **87** are formed.

(28) Still referring to FIG. **6**, after the LDD regions **65** are formed, gate spacers **87** are formed on the gate structure. In the example of FIG. **6**, the gate spacers **87** are formed on opposing sidewalls of the gate electrode **68** and on opposing sidewalls of the gate dielectric **66**. The gate spacers **87** may be formed of silicon nitride, silicon oxynitride, silicon carbide, silicon carbonitride, the like, or a combination thereof, and may be formed using, e.g., a thermal oxidation, CVD, or other suitable deposition process.

(29) The shapes and formation methods of the gate spacers **87** as illustrated in FIG. **6** are merely non-limiting examples, and other shapes and formation methods are possible. For example, the gate spacers **87** may include first gate spacers (not shown) and second gate spacers (not shown). The first gate spacers may be formed on the opposing sidewalls of the dummy gate structure **75**. The second gate spacers may be formed on the first gate spacers, with the first gate spacers disposed between a respective gate structure and the respective second gate spacers. The first gate spacers may have an L-shape in a cross-sectional view. As another example, the gate spacers **87** may be formed after the epitaxial source/drain regions **80** (see FIG. **7**) are formed. In some embodiments, dummy gate spacers are formed on the first gate spacers (not shown) before the epitaxial process of the epitaxial source/drain regions **80** illustrated in FIG. **7**, and the dummy gate spacers are removed and replaced with the second gate spacers after the epitaxial source/drain regions **80** are formed. All such embodiments are fully intended to be included in the scope of the present disclosure.

(30) Next, as illustrated in FIG. **7**, source/drain regions **80** are formed. The source/drain regions **80** are formed by etching the fins **64** to form recesses, and epitaxially growing a material in the recess, using suitable methods such as metal-organic CVD (MOCVD), molecular beam epitaxy (MBE), liquid phase epitaxy (LPE), vapor phase epitaxy (VPE), selective epitaxial growth (SEG), the like, or a combination thereof.

(31) As illustrated in FIG. **7**, the epitaxial source/drain regions **80** may have surfaces raised from respective surfaces of the fins **64** (e.g. raised above the non-recessed portions of the fins **64**) and may have facets. The source/drain regions **80** of the adjacent fins **64** may merge to form a continuous epitaxial source/drain region **80**. In some embodiments, the source/drain regions **80** of adjacent fins **64** do not merge together and remain separate source/drain regions **80**. In some

example embodiments in which the resulting FinFET is an n-type FinFET, source/drain regions **80** comprise silicon carbide (SiC), silicon phosphorous (SiP), phosphorous-doped silicon carbon (SiCP), or the like. In alternative exemplary embodiments in which the resulting FinFET is a p-type FinFET, source/drain regions **80** comprise SiGe, and a p-type impurity such as boron or indium.
(32) The epitaxial source/drain regions **80** may be implanted with dopants to form source/drain regions **80** followed by an anneal process. The implanting process may include forming and patterning masks such as a photoresist to cover the regions of the FinFET that are to be protected from the implanting process. The source/drain regions **80** may have an impurity (e.g., dopant) concentration in a range from about 1E19 cm.sup.−3 to about 1E21 cm.sup.−3. In some embodiments, the epitaxial source/drain regions may be in situ doped during growth.
(33) Next, as illustrated in FIG. **8**, a contact etch stop layer (CESL) **89** is formed over the structure illustrated in FIG. **7**. The CESL **89** functions as an etch stop layer in a subsequent etching process, and may comprise a suitable material such as silicon oxide, silicon nitride, silicon oxynitride, combinations thereof, or the like, and may be formed by a suitable formation method such as CVD, PVD, combinations thereof, or the like.
(34) Next, a first interlayer dielectric (ILD) **90** is formed over the CESL **89** and over the dummy gate structures **75**. In some embodiments, the first ILD **90** is formed of a dielectric material such as silicon oxide, phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate Glass (BPSG), undoped silicate glass (USG), or the like, and may be deposited by any suitable method, such as CVD, PECVD, or FCVD. A planarization process, such as CMP, may be performed to remove the mask **70** and to remove portions of the CESL **89** disposed over the gate electrode **68**. After the planarization process, the top surface of the first ILD **90** is level with the top surface of the gate electrode **68**.
(35) Next, in FIG. **9**, a gate-last process (sometimes referred to as replacement gate process) is performed to replace the gate electrode **68** and the gate dielectric **66** with an active gate (may also be referred to as a replacement gate or a metal gate) and active gate dielectric material(s), respectively. Therefore, the gate electrode **68** and the gate dielectric **66** may be referred to as dummy gate electrode and dummy gate dielectric, respectively, in a gate-last process. The active gate is a metal gate, in some embodiments.
(36) Referring to FIG. **9**, the dummy gate structures **75** are replaced by replacement gate structures **97** (e.g., **97**A, **97**B, **97**C). In accordance with some embodiments, to form the replacement gate structures **97**, the gate electrode **68** and the gate dielectric **66** directly under the gate electrode **68** are removed in an etching step(s), so that recesses (not shown) are formed between the gate spacers **87**. Each recess exposes the channel region of a respective fin **64**. During the dummy gate removal, the gate dielectric **66** may be used as an etch stop layer when the gate electrode **68** is etched. The gate dielectric **66** may then be removed after the removal of the gate electrode **68**.
(37) Next, a gate dielectric layer **94**, a barrier layer **96**, a work function layer **98**, and a gate electrode **95** are formed in the recesses for the replacement gate structure **97**. The gate dielectric layer **94** is deposited conformally in the recesses, such as on the top surfaces and the sidewalls of the fins **64** and on sidewalls of the gate spacers **87**, and on a top surface of the first ILD **90** (not shown). In accordance with some embodiments, the gate dielectric layer **94** comprises silicon oxide, silicon nitride, or multilayers thereof. In other embodiments, the gate dielectric layer **94** includes a high-k dielectric material, and in these embodiments, the gate dielectric layers **94** may have a k value (e.g., dielectric constant) greater than about 7.0, and may include a metal oxide or a silicate of Hf, Al, Zr, La, Mg, Ba, Ti, Pb, and combinations thereof. The formation methods of gate dielectric layer **94** may include molecular beam deposition (MBD), atomic layer deposition (ALD), PECVD, and the like.
(38) Next, the barrier layer **96** is formed conformally over the gate dielectric layer **94**. The barrier layer **96** may comprise an electrically conductive material such as titanium nitride, although other materials, such as tantalum nitride, titanium, tantalum, or the like, may alternatively be utilized.

The barrier layer **96** may be formed using a CVD process, such as PECVD. However, other alternative processes, such as sputtering, metal organic chemical vapor deposition (MOCVD), or ALD, may alternatively be used.
(39) Next, the work function layer **98**, such as a p-type work function layer or an n-type work function layer, may be formed in the recesses over the barrier layers **96** and before the gate electrode **95** is formed, in some embodiments. Exemplary p-type work function metals that may be included in the gate structures for p-type devices include TiN, TaN, Ru, Mo, Al, WN, ZrSi.sub.2, MoSi.sub.2, TaSi.sub.2, NiSi.sub.2, other suitable p-type work function materials, or combinations thereof. Exemplary n-type work function metals that may be included in the gate structures for n-type devices include Ti, Ag, TaAl, TaAlC, TiAlN, TaC, TaCN, TaSiN, Mn, Zr, other suitable n-type work function materials, or combinations thereof. A work function value is associated with the material composition of the work function layer, and thus, the material of the work function layer is chosen to tune its work function value so that a target threshold voltage Vt is achieved in the device that is to be formed. The work function layer(s) may be deposited by CVD, physical vapor deposition (PVD), and/or other suitable process.
(40) Next, a seed layer (not shown) is formed conformally over the work function layer **98**. The seed layer may include copper, titanium, tantalum, titanium nitride, tantalum nitride, the like, or a combination thereof, and may be deposited by ALD, sputtering, PVD, or the like. In some embodiments, the seed layer is a metal layer, which may be a single layer or a composite layer comprising a plurality of sub-layers formed of different materials. For example, the seed layer comprises a titanium layer and a copper layer over the titanium layer.
(41) Next, the gate electrode **95** is deposited over the seed layer, and fills the remaining portions of the recesses. The gate electrode **95** may be made of a metal-containing material such as Cu, Al, W, the like, combinations thereof, or multi-layers thereof, and may be formed by, e.g., electroplating, electroless plating, or other suitable method. After the formation of the gate electrode **95**, a planarization process, such as a CMP, may be performed to remove the excess portions of the gate dielectric layer **94**, the barrier layer **96**, the work function layer **98**, the seed layer, and the gate electrode **95**, which excess portions are over the top surface of the first ILD **90**. The resulting remaining portions of the gate dielectric layer **94**, the barrier layer **96**, the work function layer **98**, the seed layer, and the gate electrode **95** thus form the replacement gate structure **97** (also referred to as the metal gate structure) of the resulting FinFET device **100**. As illustrated in FIG. **9**, due to the planarization process, the metal gate structure **97**, the gate spacers **87**, the CESL **89**, and the first ILD **90** have a coplanar upper surface.
(42) Next, in FIG. **10**, a metal gate etch-back process is performed to remove upper portions of the metal gate structures **97**, such that the metal gate structures **97** recess below the upper surface of the first ILD **90**. Recesses **88** are formed between the gate spacers **87** after the metal gate etch-back process. A suitable etching process, such as dry etch, wet etch, or combinations thereof, may be performed as the metal gate etch-back process. An etchant for the etching process may be a halide (e.g., CCl.sub.4), an oxidant (e.g., O.sub.2), an acid (e.g., HF), a base (e.g., NH.sub.3), an inert gas (e.g., Ar), combinations thereof, or the like, as examples.
(43) Next, in FIG. **11**, the gate spacers **87** and the CESL **89** are recessed below the upper surface of the first ILD **90**. In some embodiments, an anisotropic etching process, such as a dry etch process, is performed to remove upper portions of the gate spacer **87**. The CESL **89** may be removed by the same anisotropic etching process, if the CESL **89** and the gate spacers **87** comprise the same material, or have a same or similar etch rate for the anisotropic etching process. In some embodiments, the anisotropic etching process is performed using an etchant that is selective to (e.g., having a higher etch rate for) the material(s) of the gate spacers **87**/CESL **89**, such that the gate spacers **87**/CESL **89** are recessed (e.g., upper portions removed) without substantially attacking the first ILD **90** and the metal gate structures **97**. In embodiments where the gate spacers **87** and the CESL **89** have different etch rates, a first anisotropic etching process using a first

etchant selective to the material of the gate spacers **87** may be performed to recess the gate spacers **87**, and a second anisotropic etching process using a second etchant selective to the material of the CESL **89** may be performed to recess the CESL **89**. Upper surfaces of the recessed gate spacers **87** and upper surfaces of the recessed CESL **89** may be level with the respective upper surfaces of the metal gate structures **97**. In some embodiments, the CESL **89** is recessed after the capping layer **91** (discussed below) is formed.

(44) Next, the metal gate structures **97** are recessed again, e.g., using the same or similar metal gate etch-back process discussed above, such that the upper surfaces of the metal gate structures **97** are lower (e.g., closer to the substrate) than the upper surfaces of the gate spacers **87**. Next, a capping layer **91** is formed on the upper surfaces of the metal gate structures **97** to protect the metal gates structures **97**, e.g., from oxidization and/or subsequent etching processes. The capping layer **91** is formed of a conductive material (e.g., metal), and is formed selectively on the upper surfaces of the metal gate structures **97**, in the illustrated example. The capping layer **91** may be formed of, e.g., tungsten, although other suitable conductive material may also be used. A suitable formation method, such as CVD, PVD, ALD, or the like, may be used to form the capping layer **91**. Note that in the discussion herein, unless otherwise specified, a conductive material refers to an electrically conductive material, and a conductive feature (e.g., a conductive line) refers to an electrically conductive feature.

(45) In the example of FIG. **11**, the capping layer **91**, the recessed gate spacers **87**, and the recessed CESL **89** have a level (e.g., coplanar) upper surface. In other embodiments, there are offsets (e.g., vertical distances) among the upper surfaces of the capping layer **91**, the recessed gate spacers **87**, and the recessed CESL **89**. Due to the recessing of the gate spacers **87** and the CESL **89**, the recesses **88** in FIG. **10** are expanded and are denoted as recesses **88′** in FIG. **11**.

(46) Next, in FIG. **12**, a dielectric material **99** is formed to fill the recesses **88′**, and a planarization process, such as CMP, may be performed next to remove excess portions of the dielectric material **99** from the upper surface of the first ILD **90**. In an embodiment, the dielectric material **99** is a nitride (e.g., silicon nitride, silicon oxynitride, silicon carbonitride). The dielectric material **99** may be formed using any suitable formation method such as CVD, PECVD, or the like. The dielectric material **99** protects the underlying structures, such as the metal gate structure **97**, the gate spacers **87**, and portions of the underlying CESL **89** from a subsequent etching process(es) for forming source/drain contacts. Details are discussed hereinafter.

(47) Next, in FIG. **13**, a dielectric layer **101** is formed over the first ILD **90**, and a patterned mask layer **102**, such as a patterned photoresist, is formed over the dielectric layer **101**. The dielectric layer **101** may comprise a same or similar material as the first ILD **90** and may be formed of a same or similar formation method as the first ILD **90**, thus details are not repeated. In the example of FIG. **13**, an opening in the patterned mask layer **102** is over (e.g., directly over) some of the source/drain regions **80** and over (at least portions of) the dielectric material **99**.

(48) Next, an etching process is performed to remove portions of the first ILD **90** and portions of the dielectric layer **101** that underlie the opening of the patterned mask layer **102**. The etching process may be an anisotropic etching process, such as a reactive ion etch (RIE), an atomic layer etch (ALE), or the like. The etching process may use an etchant that is selective to (e.g., having a higher etch rate for) the material(s) of the first ILD **90** and the dielectric layer **101**. As illustrated in FIG. **13**, after the etching process, openings **104** are formed in the first ILD **90**, such as between opposing sidewalls of the CESL **89** and over source/drain regions **80**. The openings **104** expose the underlying source/drain regions **80**. The openings **104** are used to form self-aligned source/drain contacts **109** (see FIG. **14**) in subsequently processing. The number and the locations of the openings **104** in FIG. **13** are merely non-limiting examples, one skilled in the art will readily appreciate that any numbers of the openings **104** may be formed, and the locations of the openings **104** may be at any suitable locations.

(49) The dielectric material **99** protects (e.g., shields) the underlying structures, such as the gate

spacers **87** and the CESL **89**, from the anisotropic etching process to form the openings **104**. It is observed that during manufacturing, corner regions **107** of the gate spacers **87**/CESL **89** tend to be etched away at a faster rate than other regions of the gate spacers **87**/CESL **89**, resulting in the "shoulder loss" problem. The shoulder loss problem may be caused by the decreased etching selectivity between the material(s) of the gate spacers **87**/CESL **89** and the material(s) of the first ILD **90**/dielectric layer **101**, which decreased etching selectivity may be a result of decreasing critical dimension (CD) in advanced semiconductor manufacturing. If the shoulder loss problem cause the metal gate structure **97** to be exposed, electrical short between the metal gate structure **97** and the adjacent source/drain region **80** may occur, when the openings **104** are filled with a conductive material in subsequent processing. The dielectric material **99** shields the gate spacers **87**/CESL **89** from the anisotropic etching process, thus reducing or preventing occurrence of the shoulder loss, which in turn reduces or prevents device failure caused by electrical short between the metal gate structures **97** and the source/drain regions **80**.

(50) Next, in FIG. **14**, a barrier layer **105** is formed conformally over the structure of FIG. **13**. The barrier layer **105** may comprise titanium, tantalum, titanium nitride, tantalum nitride, or the like, and may be formed using a suitable formation method such as ALD, CVD, or the like. In some embodiments, the barrier layer **105** is formed to line sidewalls and bottoms of the openings **104**.

(51) Next, silicide regions **108** are formed over the source/drain regions **80** exposed by the openings **104**. The silicide regions **108** may be formed by first depositing a metal layer capable of reacting with semiconductor materials (e.g., silicon, germanium) to form silicide or germanide regions, such as nickel, cobalt, titanium, tantalum, platinum, tungsten, other noble metals, other refractory metals, rare earth metals or their alloys, over the source/drain regions **80**, then performing a thermal anneal process to form the silicide regions **108**. In some embodiments, the un-reacted portions of the deposited metal layer are removed, e.g., by an etching process after the thermal anneal process. Although regions **108** are referred to as silicide regions, the regions **108** may also be germanide regions, or silicon germanide regions (e.g., regions comprising silicide and germanide). In an example embodiment where the barrier layer **105** comprises a suitable metal material such as titanium or tantalum, the silicide regions **108** are formed by performing a thermal anneal process after the barrier layer **105** is formed, such that portions of the barrier layer **105** at the bottoms of the openings **104** (e.g., on the source/drain regions **80**) react with the source/drain regions **80** to form silicide regions **108**.

(52) Next, a conductive material, such as titanium, cobalt, or the like, is formed to fill the openings **104** using a suitable formation methods, such as PVD, CVD, ALD, plating, and the like. Next, a planarization process, such as CMP, is performed to remove the patterned mask layer **102**, the dielectric layer **101**, and portions of the conductive material disposed outside the openings **104**. The remaining portions of the conductive material in the openings **104** form self-aligned source/drain contacts **109**. For simplicity, the self-aligned source/drain contacts **109** may also be referred to as source/drain contacts **109**.

(53) FIG. **14** further illustrates a metal region **103** over, e.g., the upper surface of dielectric material **99**. The metal region **103** (may also be referred to as a metal layer) is a residue portion of the conductive material of the source/drain contacts **109** that is not removed by the planarization process. In other words, the metal region **103** and the source/drain contacts **109** are formed of a same conductive material, in some embodiments. The metal region **103** may be formed due to, e.g., the upper surface of the dielectric material **99** not being perfectly flat (e.g., having divots), and therefore, the conductive material of the source/drain contacts **109** is deposited into the divots and was not removed by the planarization process. The metal region **103** ideally should not exist. However, if the planarization process is insufficient and the metal region **103** remain on the dielectric material **99**, a subsequent etching process to form via holes through the dielectric material **99** may be stopped prematurely by the metal region **103**, thereby causing device failure. The present disclosure discloses a multi-step etching process to ensure that via holes will be formed

properly regardless of whether the metal region **103** exists. Details are discussed hereinafter. Note that the number and the location of the metal region **103** in FIG. **14** are for illustration purpose only and not limiting.

(54) Next, in FIG. **15**, an etch stop layer **111**, a dielectric layer **112**, and a tri-layered photoresist **116** are formed successively over the first ILD **90**. The etch stop layer **111** is formed of suitable material, e.g., silicon nitride, silicon carbide, silicon carbonitride, or the like, by a suitable formation methods such as CVD, PECVD, ALD, or the like. The dielectric layer **112** may comprise a same or similar material as the first ILD **90**, and may be formed of a same or similar formation method, thus details are not repeated.

(55) In some embodiments, the tri-layered photoresist **116** includes a top photoresist layer **117**, a middle layer **115**, and a bottom anti-reflective coating (BARC) layer **113**. The BARC layer **113** of the tri-layered photoresist **116** may comprise an organic or inorganic material. The middle layer **115** may comprise silicon nitride, silicon oxynitride, or the like, that has an etch selectivity to the top photoresist layer **117**, such that the top photoresist layer **117** can be used as a mask layer to pattern the middle layer **115**. The top photoresist layer **117** may comprise a photosensitive material. Any suitable deposition method, such as PVD, CVD, spin coating, the like, or combinations thereof, may be used to form the tri-layered photoresist **116**.

(56) Once the tri-layered photoresist **116** is formed, patterns **118** (also referred to as openings **118**) are formed in the top photoresist layer **117**, e.g., using photolithography and etching techniques. The patterns **118** are formed over (e.g., directly over) metal gate structures **97**, in the illustrated embodiment.

(57) Next, in FIG. **16**, the patterns **118** (e.g., **118**A, **118**C) in the top photoresist layer **117** are extended through the middle layer **115** and the BARC layer **113**, and are transferred to the underlying layers (e.g., the dielectric layer **112**, the etch stop layer **111**, the dielectric material **99**) using an anisotropic etching processes, such as a first dry etch process. In the example of FIG. **16**, the opening **118**A over the metal gate structure **97**A extends through the dielectric layer **112**, through the etch stop layer **111**, and into the dielectric material **99**. In other words, after the first dry etch process, the bottom of the opening **118**A is between the upper surface and the lower surface of the dielectric material **99**. Note that the opening **118**C over the metal gate structure **97**C extends through the dielectric layer **112**, through the etch stop layer **111**, but stops at (e.g., exposes) the metal region **103**, due to, e.g., the etching selectivity between the dielectric layer **112**/etch stop layer **111** and the metal region **103**. In other words, the metal region **103** prevents the first dry etch process from reaching a target depth for the opening **118**C.

(58) In some embodiments, the first dry etch process is a first plasma process (also referred to as a plasma etch process) performed using a process gas comprising C.sub.4F.sub.6, C.sub.4F.sub.8, CH.sub.2F.sub.2, or combinations thereof. A carrier gas, such as N.sub.2 or He, may be used to carry the process gas into the process chamber. In accordance with some embodiments, the first plasma process is a direct plasma process, with the plasma being generated in the same process chamber where the FinFET device **100** is treated. The first plasma process is performed using both a High-Frequency Radio-Frequency (HFRF) power (e.g., with a frequency about 60 MHz) and a Low-Frequency Radio-Frequency (LFRF) power (e.g., with a frequency of about 2 MHz). The HFRF power is used for ionization and to generate plasma, and the LFRF power (also referred to as bias power) is used for bombarding the layers (e.g., **112**, **111**, and **99**) for removal. In accordance with some embodiments of the present disclosure, the HFRF power of the first plasma process is in the range between about 50 watts and about 400 watts, and the LFRF power of the first plasma process is between about 400 watts and about 600 watts. In some embodiments, the range of the LFRF power (e.g., between 400 watts and about 600 watts) is chosen to provide a target level of etching capability (e.g., etch rate) for the dielectric layer **112**, to maintain a target level of etching selectivity between the dielectric layer **112** (e.g., silicon oxide) and the dielectric material **99** (e.g., silicon nitride), and to avoid under-etch during the first dry etch process.

(59) Next, in FIG. **17**, a wet etch process (also referred to as a wet cleaning process) is performed to clean the openings **118**. In some embodiments, the wet etch process is performed using a wet clean chemical, which may be, e.g., a mixture of HCl and H.sub.2CO.sub.3 dissolved in water. The wet etch process removes residues and by-products (e.g., polymer) of the first dry etch process from the openings **118**. The wet etch process also etches through the metal region **103** at the bottom of the opening **118**C to expose the underlying dielectric material **99**. In some embodiments, the wet cleaning chemical is selective to (e.g., having a higher etch rate for) the residues/by-products (e.g., polymer) of the first dry etch process and the metal region **103**, such that the residues/by-products and the metal region **103** are removed (e.g., etched) without substantially attacking other layers/materials.
(60) Next, in FIG. **18**, a second dry etch process is performed to extend the openings **118** through the dielectric material **99** and to expose the capping layer **91**. In some embodiments, the second dry etch process is a second plasma process performed using a process gas comprising CH.sub.2F.sub.2 and H.sub.2. In some embodiments, the second plasma process is similar to the first plasma process of the first dry etch process, and the HFRF power of the second plasma process is in the range between about 50 watts and about 400 watts, and the LFRF power of the second plasma process is between about 0 watts and about 150 watts. In some embodiments, the LFRF power (e.g., less than about 150 watts) of the second plasma process is chosen to be smaller than the LFRF power of the first plasma process. The LFRF power of the second plasma process is chosen to provide a target level of etching capability (e.g., etch rate) for the dielectric material **99**, to achieve a more uniform depth of the openings **118** in the dielectric material **99**, and to avoid enlargement of the size (e.g., width W) of the openings **118**. As illustrated in FIGS. **16-18**, the multi-step etching process, which includes the first dry etch process, the wet etch process, and the second dry etch process, ensures that via holes (e.g., **118**) are formed properly regardless of whether the metal region **103** exists.
(61) Next, in FIG. **19**, the tri-layered photoresist **116** is removed (e.g., by an ashing process). Next, a conductive material **121** is formed to fill the openings **118**. The conductive material may be, e.g., tungsten, titanium, or the like, and may be formed by a suitable formation method such as CVD, PECVD, ALD, or the like. Next, a planarization process, such as CMP, is performed to remove the dielectric layer **112**, and portions of the conductive material **121** over the upper surface of the etch stop layer **111**. In other words, the planarization process stops after the etch stop layer **111** is exposed. The remaining portions of the conductive material **121** form vias **121** that are over and electrically coupled to the metal gate structures **97**. FIG. **19** further illustrate a metal region **124** on the upper surface of the etch stop layer **111**. The metal region **124** may comprise residue portions of the conductive material **121** that are grinded away by the CMP process and deposited in a lower area at the upper surface of the etch stop layer **111**. Note that the number and the location of metal region **124** in FIG. **19** are for illustration purpose only and not limiting
(62) The metal region **124** in FIG. **19** is directly over the source/drain contact **109**. The metal region **124** may cause a subsequent etching process to form via holes (see, e.g., **128** in FIG. **21**) to stop prematurely. The multi-step etching process same as or similar to those discussed above may be used to form vias holes properly, regardless of whether the metal region **124** exists. Details are discussed hereinafter.
(63) Next, in FIG. **20**, a dielectric layer **122** and a tri-layered photoresist **126** are formed successively over the etch stop layer **111**. The dielectric layer **122** and the tri-layered photoresist **126** may be same as or similar to the dielectric layer **112** and the tri-layered photoresist **116** of FIG. **18**, thus details are not repeated.
(64) As illustrated in FIG. **20**, patterns **128** (also referred to as openings **128**) are formed in the top photoresist layer **127** of the tri-layered photoresist **126**, e.g., using photolithography and etching techniques. The patterns **128** are formed over (e.g., directly over) source/drain contacts **109**, in the illustrated embodiment.
(65) Next, in FIG. **21**, the patterns **128** (e.g., **128**A, **128**B) in the top photoresist layer **127** are

extended through the middle layer **125** and the BARC layer **123** of the tri-layered photoresist **126**, and are transferred to the underlying layers (e.g., the dielectric layer **122**, the etch stop layer **111**) using the first dry etch process discussed above. In the example of FIG. **21**, the opening **128**B extends through the dielectric layer **122** and into the etch stop layer **111**. In other words, after the first dry etch process, the bottom of the opening **128**B is between the upper surface and the lower surface of the etch stop layer **111**. Note that the opening **128**A extends through the dielectric layer **122**, but stops at (e.g., exposes) the metal region **124**, due to, e.g., the etching selectivity between the dielectric layer **122** and the metal region **124**.

(66) Next, in FIG. **22**, the wet etch process described above is performed to clean the openings **128**. The wet etch process removes residues and by-products (e.g., polymer) of the first dry etch process. The wet etch process also etches through the metal region **124** at the bottom of the opening **128**A to expose the etch stop layer **111**. In some embodiments, the wet cleaning chemical is selective to (e.g., having a higher etch rate for) the residues/by-products (e.g., polymer) of the first dry etch process and the metal region **124**, such that the residues/by-products and the metal region **124** are removed (e.g., etched) without substantially attacking other layers/materials.

(67) Next, in FIG. **23**, the second dry etch process described above is performed to extend the openings **128** through the etch stop layer **111** and to expose the source/drain contacts **109**. As illustrated in FIGS. **21-23**, the multi-step etching process, which includes the first dry etch process, the wet etch process, and the second dry etch process, ensures that via holes (e.g., **128**) are formed properly regardless of whether the metal region **124** exists.

(68) Next, in FIG. **24**, a conductive material **131** is formed to fill the openings **128**. The conductive material may be, e.g., tungsten, titanium, or the like, and may be formed by a suitable formation method such as CVD, PECVD, ALD, or the like. Next, a planarization process, such as CMP, is performed to remove the tri-layered photoresist **126**, the dielectric layer **122**, and portions of the conductive material **131** over the upper surface of the etch stop layer **111**. The remaining portions of the conductive material **131** form vias **131** that are over and electrically coupled to the underlying source/drain contacts **109**.

(69) Additional processing may be performed after the processing of FIG. **24** to finish fabrication of the FinFET device **100**. For example, an interconnect structure, which includes multiples dielectric layers and conductive features (e.g., vias, conductive lines) formed in the multiple dielectric layers, is formed over the structure of FIG. **24** to interconnect the underlying electrical components to form functional circuits. Details are not discussed here.

(70) Embodiments may achieve advantages. For example, the multi-step etching process disclosed herein, which includes the first dry etch process, the wet etch process, and the second dry etch process, ensures that via holes (e.g., **118**, **128**) are formed properly regardless of whether the metal regions **103** or **124** exist. As a result, product defects caused by insufficient via hole etching is avoided or reduced.

(71) FIG. **25** illustrates a flow chart of a method **1000** of fabricating a semiconductor device, in accordance with some embodiments. It should be understood that the embodiment method shown in FIG. **25** is merely an example of many possible embodiment methods. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. For example, various steps as illustrated in FIG. **25** may be added, removed, replaced, rearranged and repeated.

(72) Referring to FIG. **25**, at block **1010**, a gate structure is recessed below an upper surface of a dielectric layer that surrounds the gate structure. At block **1020**, a first dielectric material is formed over the recessed gate structure. At block **1030**, a second dielectric material is formed over the first dielectric material. At block **1040**, an opening is formed that extends through the second dielectric material and into the first dielectric material using a first dry etch process. At block **1050**, after the first dry etch process, the opening is cleaned by a wet cleaning process. At block **1060**, after the wet cleaning process, the opening is extended through the first dielectric material using a second dry etch process. At block **1070**, the opening is filled with a conductive material.

(73) In an embodiment, a method of forming a semiconductor device includes: forming a metal gate structure over a fin, the fin protruding above a substrate, the metal gate structure being surrounded by an interlayer dielectric (ILD) layer; recessing the metal gate structure below an upper surface of the ILD layer distal from the substrate; after the recessing, forming a first dielectric layer over the recessed metal gate structure; forming an etch stop layer over the first dielectric layer and the ILD layer; forming a second dielectric layer over the etch stop layer; performing a first dry etch process to form a first opening that extends through the second dielectric layer, through the etch stop layer, and into the first dielectric layer; after the first dry etch process, performing a wet etch process to clean the first opening; and after the wet etch process, performing a second dry etch process to extend the first opening through the first dielectric layer. In an embodiment, the method further includes, after the second dry etch process, filling the first opening with a conductive material. In an embodiment, an upper surface of the first dielectric layer is formed to be level with the upper surface of the ILD layer. In an embodiment, the method further includes, before forming the first dielectric layer, recessing gate spacers of the metal gate structure below the upper surface of the ILD layer, wherein the first dielectric layer is formed over the recessed gate spacers. In an embodiment, the first dry etch process is a first plasma process, and the second dry etch process is a second plasma process. In an embodiment the first plasma process is performed with a first bias power, and the second plasma process is performed with a second bias power different from the first bias power. In an embodiment, the second bias power is lower than the first bias power. In an embodiment, the first dielectric layer is formed of silicon nitride, and the second dielectric layer is formed of silicon oxide. In an embodiment, the first dry etch process is a first plasma process performed using a first process gas comprising C.sub.4F.sub.6, C.sub.4F.sub.8, or CH.sub.2F.sub.2, and the second dry etch process is a second plasma process performed using a second process gas comprising CH.sub.2F.sub.2 and H.sub.2. In an embodiment, the wet etch process is performed using a wet clean chemical comprising HCl and H.sub.2CO.sub.3. In an embodiment, the method further includes, after forming the first dielectric layer and before forming the etch stop layer: removing a portion of the ILD layer adjacent to the metal gate structure to form a second opening in the ILD layer, the second opening exposing an underlying source/drain region; filling the second opening with a conductive material; and after the filling, performing a planarization process to remove portions of the conductive material disposed outside the second opening. In an embodiment, after the planarization process, a residue portion of the conductive material remains on an upper surface of the first dielectric layer, wherein the method further comprises: forming a third opening that extends through the second dielectric layer and the etch stop layer using the first dry etch process, wherein a bottom of the third opening exposes the residue portion of the conductive material; etching through the residue portion of the conductive material to expose the first dielectric layer using the wet etch process; and extending the third opening through the first dielectric layer using the second dry etch process.
(74) In an embodiment, a method of forming a semiconductor device includes: recessing a gate structure below an upper surface of a dielectric layer, the dielectric layer surrounding the gate structure; forming a first dielectric material over the recessed gate structure; forming a second dielectric material over the first dielectric material; forming an opening that extends through the second dielectric material and into the first dielectric material using a first dry etch process; after the first dry etch process, wet cleaning the opening; after the wet cleaning, extending the opening through the first dielectric material using a second dry etch process; and filling the opening with a conductive material. In an embodiment, the first dry etch process is a first plasma process, and the second dry etch process is a second plasma process. In an embodiment, a first bias power of the first plasma process is higher than a second bias power of the second plasma process. In an embodiment, the first dielectric material is a nitride, and the second dielectric material is an oxide. In an embodiment, the first plasma process is performed using a first process gas comprising C.sub.4F.sub.6, C.sub.4F.sub.8, or CH.sub.2F.sub.2, the second plasma process is performed using

a second process gas comprising CH_2F_2 and H_2, and the wet cleaning is performed using a wet clean chemical comprising HCl and H_2CO_3.
(75) In an embodiment, a method of forming a semiconductor device includes: forming a gate structure over a fin that protrudes above a substrate, wherein the gate structure is surrounded by a dielectric layer; forming a nitride layer over the gate structure; forming an etch stop layer over the nitride layer; forming an oxide layer over the etch stop layer; forming a first opening that extends through the oxide layer, through the etch stop layer, and into the nitride layer by a first dry etch process; cleaning the first opening by a wet etch process after the first dry etch process; after cleaning the first opening, extending the first opening through the nitride layer by a second dry etch process; and filling the first opening with a conductive material after the second dry etch process. In an embodiment, the first dry etch process is a first plasma etch process performed with a first bias power, and the second dry etch process is a second plasma etch process performed with a second bias power lower than the first bias power. In an embodiment, the method further includes: forming a second opening that extends through the oxide layer and through the etch stop layer by the first dry etch process, wherein a bottom of the second opening exposes a metal area disposed between the etch stop layer and the nitride layer; etching through the metal area by the wet etch process; and extending the second opening through the nitride layer by the second dry etch process.
(76) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method of forming a semiconductor device, the method comprising: forming a gate structure over a fin, the fin protruding above a substrate, the gate structure being surrounded by an interlayer dielectric (ILD) layer; recessing the gate structure below an upper surface of the ILD layer distal from the substrate; after the recessing, forming a first dielectric layer over the recessed gate structure; forming an etch stop layer over the first dielectric layer and the ILD layer; forming a second dielectric layer over the etch stop layer; performing a first plasma etch process to form a first opening that extends through the second dielectric layer, through the etch stop layer, and into the first dielectric layer; after the first plasma etch process, performing a wet etch process to clean the first opening; and after the wet etch process, performing a second plasma etch process to extend the first opening through the first dielectric layer.
2. The method of claim 1, wherein forming the gate structure comprises: forming a dummy gate structure over the fin; forming the ILD layer over the fin around the dummy gate structure; and replacing the dummy gate structure with the gate structure, comprising: removing the dummy gate structure to form a recess in the ILD layer; and forming the gate structure in the recess over the fin.
3. The method of claim 1, further comprising, after the second plasma etch process, forming a via by filling the first opening with a conductive material.
4. The method of claim 1, wherein an upper surface of the first dielectric layer is formed to be level with the upper surface of the ILD layer.
5. The method of claim 4, further comprising, after recessing the gate structure and before forming the first dielectric layer: recessing gate spacers of the gate structure below the upper surface of the ILD layer; and forming a capping layer between the recessed gate spacers over the recessed gate structure, wherein the capping layer is formed of a conductive material, wherein the first dielectric

layer is formed over the recessed gate spacers and the capping layer.
6. The method of claim 1, wherein the first plasma etch process is performed with a first bias power, and the second plasma etch process is performed with a second bias power different from the first bias power.
7. The method of claim 6, wherein the second bias power is lower than the first bias power.
8. The method of claim 1, wherein the first dielectric layer is formed of a nitride material, and the second dielectric layer is formed of an oxide material.
9. The method of claim 8, wherein the first plasma etch process is performed using a first process gas comprising C.sub.4F.sub.6, C.sub.4F.sub.8, or CH.sub.2F.sub.2, and the second plasma etch process is performed using a second process gas comprising CH.sub.2F.sub.2 and H.sub.2.
10. The method of claim 9, wherein the wet etch process is performed using a chemical comprising HCl and H.sub.2CO.sub.3.
11. The method of claim 1, further comprising, after forming the first dielectric layer and before forming the etch stop layer: forming a second opening in the ILD layer adjacent to the gate structure, wherein the second opening exposes an underlying source/drain region; filling the second opening with a conductive material; and after the filling, performing a planarization process to remove portions of the conductive material disposed outside the second opening.
12. The method of claim 11, wherein after the planarization process, a residue portion of the conductive material remains on an upper surface of the first dielectric layer, wherein the method further comprises: forming a third opening that extends through the second dielectric layer and the etch stop layer using the first plasma etch process, wherein a bottom of the third opening exposes the residue portion of the conductive material; etching through the residue portion of the conductive material to expose the first dielectric layer using the wet etch process; and extending the third opening through the first dielectric layer using the second plasma etch process.
13. A method of forming a semiconductor device, the method comprising: forming a first dielectric layer over a gate structure; forming a second dielectric layer over the first dielectric layer; forming an opening that extends through the second dielectric layer and into the first dielectric layer using a first dry etch process; after the first dry etch process, wet cleaning the opening; after the wet cleaning, extending the opening through the first dielectric layer using a second dry etch process; and after extending the opening, filling the opening with a conductive material.
14. The method of claim 13, wherein the first dry etch process is a first plasma process, and the second dry etch process is a second plasma process.
15. The method of claim 14, wherein a first bias power of the first plasma process is higher than a second bias power of the second plasma process.
16. The method of claim 14, wherein the first dielectric layer is a nitride, and the second dielectric layer is an oxide, wherein the first plasma process is performed using a first process gas comprising C.sub.4F.sub.6, C.sub.4F.sub.8, or CH.sub.2F.sub.2, the second plasma process is performed using a second process gas comprising CH.sub.2F.sub.2 and H.sub.2, and the wet cleaning is performed using a wet clean chemical comprising HCl and H.sub.2CO.sub.3.
17. The method of claim 16, wherein the gate structure is surrounded by an interlayer dielectric (ILD) layer, wherein the method further comprises recessing the gate structure before forming the first dielectric layer, wherein the first dielectric layer is formed to have a coplanar upper surface with the ILD layer.
18. A method of forming a semiconductor device, the method comprising: forming a gate structure over a fin, wherein the fin protrudes above a substrate, wherein the gate structure is surrounded by an interlayer dielectric (ILD) layer; recessing the gate structure below an upper surface of the ILD layer; forming a first dielectric layer over the recessed gate structure; forming a second dielectric layer over the first dielectric layer; forming a first opening that extends through the second dielectric layer and into the first dielectric layer by performing a first dry etch process; after performing the first dry etch process, cleaning the first opening by a wet etch process; after

cleaning the first opening, extending the first opening through the first dielectric layer by performing a second dry etch process; and filling the first opening with a conductive material after performing the second dry etch process.

19. The method of claim 18, wherein the first dry etch process is a first plasma etch process performed with a first bias power, and the second dry etch process is a second plasma etch process performed with a second bias power lower than the first bias power.

20. The method of claim 18, further comprising: forming a second opening that extends through the second dielectric layer by the first dry etch process, wherein a bottom of the second opening exposes a metal area disposed between the second dielectric layer and the first dielectric layer; etching through the metal area by the wet etch process; and extending the second opening through the first dielectric layer by the second dry etch process.